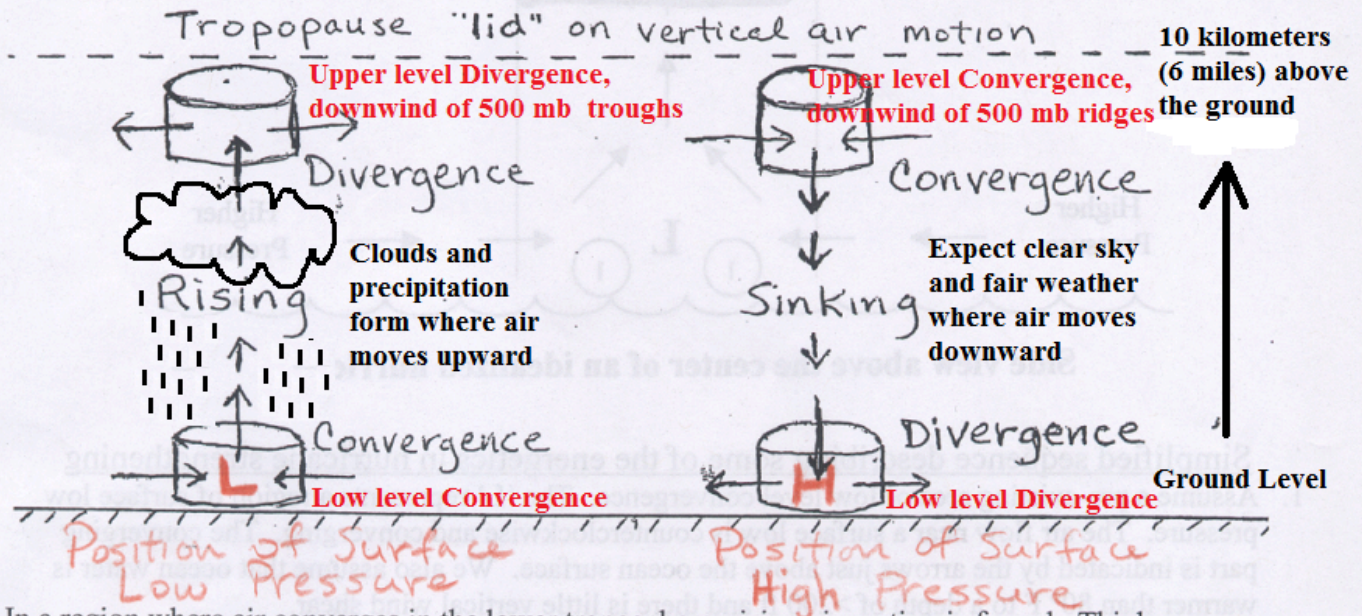


Notes on dynamical lifting of air

When we talk about **wind**, we are talking about horizontal air motion. In the atmosphere, horizontal air motion is typically much stronger than vertical air motion. This should fit with your own experience, i.e., how often can you feel air moving up or down? Under the right conditions, the horizontal wind pattern can force air to rise (or sink).

Convergence is horizontal inflow of air into a region.

Divergence is horizontal outflow of air away from a region.



In a region where air converges in the lower troposphere, rising air motion is forced. Think about the picture above. Squeezing air into the cylinder just above the ground forces air to rise because it cannot go down through the ground. This is just like squeezing a tube of toothpaste from the bottom, forcing fluid to rise. One place this happens in the atmosphere is near areas of surface low pressure (often marked with an 'L' on surface weather maps). On the other hand, in regions where air diverges in the lower troposphere, sinking air motion is forced. Using the picture above, as air is evacuated from the cylinder, other air must move in to fill the void. Since air cannot rise up through the ground, it must sink down from above. One place this commonly happens is near areas of surface high pressure (marked with an 'H' on surface weather maps).

The relationship between convergence and divergence in the upper troposphere and the resulting rising or sinking motion is reversed. To understand how this works you must take into consideration that the tropopause acts like a "lid" on top of the troposphere, i.e., it is difficult for air to move up or down through this boundary. Hopefully, you can now see that where air diverges in the upper troposphere, the air that comes in to take its place must come from below (not through the tropopause "lid"), thus rising motion is forced. Upper tropospheric divergence is forced just downwind of 500 mb troughs. Conversely, where air converges in the upper troposphere, sinking air motion is forced because the air will not move upward through the tropopause "lid". Upper tropospheric convergence is forced downwind of 500 mb ridges.

In summary, rising air motion is forced by convergence in the lower troposphere and/or divergence in the upper troposphere. Often they work together, air converges in the lower troposphere, rises upward, and then diverges in the upper troposphere.